WAIMLAp 2026

Workshop on Artificial Intelligence and Machine Learning Applications

PUCP

How Do Web-Pretrained DINOv3 Features Extend to Martian Terrain? Appearance, Adaptation and Errors on AI4Mars

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MOTIVATION & RESEARCH PROBLEM

- Foundation vision models are pretrained on web images, yet they are used on scenes that look nothing like the web, such as the surface of Mars.
- Rovers must tell soil, bedrock, sand and big rock apart, and expert labels are scarce and ambiguous.
- **Question:** how do the features of a DINOv3 pretrained on web images extend to the Martian domain (AI4Mars)?

METHODOLOGY

- **Data:** AI4Mars, Curiosity NavCam, 512 px, 4 classes. 15,564 training images, 500 validation (temporal hold-out), 322 expert test images in three sets M1 / M2 / M3 (labels agreed by at least 1 / 2 / 3 experts).
- **Backbone:** DINOv3 ViT-B/16, frozen (85.7 M parameters, 215 GFLOPs per image, 12 identical blocks of 7.1 M). Linear probe on the last layer: 4.6 k trainable parameters.
- **P1:** retrain the probe on images changed only in appearance: histogram equalisation, CLAHE, per-image standardisation, Gaussian blur.
- **P2:** LoRA rank 8 (221 k parameters) on two consecutive blocks: early (1–2), middle (6–7) or late (11–12); same budget in each zone.
- **P3:** mIoU where three experts agree vs where only one labelled; calibration (ECE) overall and on pixels predicted big rock.
- **Baseline:** DeepLabV3-R50 end to end, same split and budget (3 epochs), with and without ImageNet pretraining.
- **Statistics:** 3 seeds (2 for LoRA, 1 for the baseline); paired block-bootstrap 95 % CI; real if the CI excludes 0 and $|\Delta| \geq 0.01$, equivalent if inside ±0.01. Three hypotheses, one per question, written down before the runs.

P1: THE GAP IS NOT ABOUT APPEARANCE

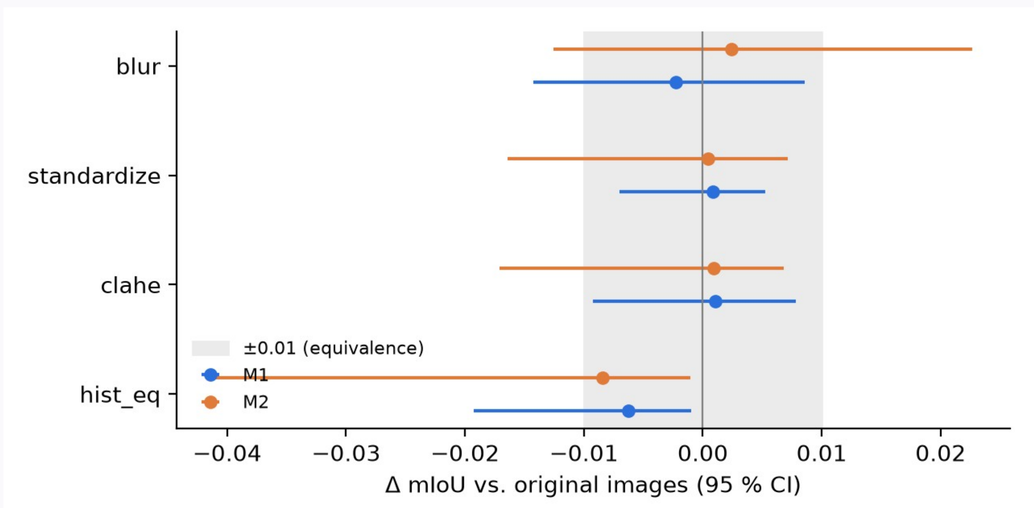


Fig. 1. P1: change in probe mIoU under each appearance transform (grey band: equivalence zone ±0.01).

CONCLUSIONS, LIMITATIONS & ETHICS

- **Web-pretrained DINOv3 features already work on Mars with almost no adaptation:** a 4.6 k-parameter probe matches an end-to-end DeepLabV3 within the statistical margin.
- The gap is not one of image appearance, and adapting early, middle or late blocks adds only about 0.006 mIoU.
- What limits the score is label ambiguity and the rare big-rock class, on which the model is confidently wrong. Better labels and class-aware losses are the next step, not a heavier model.
- **Limits:** one backbone and one rover; 322 test images, so P1 and P2 cannot be decided at this sample size (only P3 is supported); LoRA 2 seeds and 2 epochs; baseline 1 seed and 3 epochs; M3 holds big rock in only 5 images.
- **Ethics:** AI4Mars is public NASA data; labels are crowd-sourced and noisy; these models should not be relied on for hazard detection.

KEY REFERENCES

[1] O. Siméoni et al., “DINOv3,” arXiv:2508.10104, 2025.

[2] R. M. Swan et al., “AI4MARS: A dataset for terrain-aware autonomous driving on Mars,” CVPR Workshops, 2021.

[3] F. Mohammad, Y. Li, Y. Gao, “Transformers vs CNNs: Enhancing semantic segmentation for Martian terrain classification,” iSpaRo, 2025.

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[5] P. Durai, G. Doran, “MANTLE: a framework for adaptive in-situ planetary perception using a modular uplink principle,” arXiv:2608.28724, 2026.

[6] E. J. Hu et al., “LoRA: Low-rank adaptation of large language models,” ICLR, 2022.

RESEARCH QUESTIONS

- **P1, data:** is the gap one of image appearance? Change only how the photos look and watch a linear probe.
- **P2, model:** where is it worth adapting? Same LoRA budget on early, middle or late blocks.
- **P3, errors:** where does it fail and does it know? Agreement between experts, and calibration.
- Closest work uses frozen DINOv2 on other Mars data and tasks [5]; we found no controlled study of adaptation depth or calibration on AI4Mars.

RESULTS & DISCUSSION

- **Reference:** the probe reaches mIoU 0.674 / 0.723 on test M1 / M2, level with DeepLabV3-R50 (0.667 / 0.713; M1 difference −0.007, 95 % CI [−0.025, +0.008]) with about 8,600 times fewer trainable parameters. Without pretraining the baseline is clearly behind (0.628).
- **P1, appearance:** CLAHE and standardisation are equivalent to the original images on M1 (+0.001); histogram equalisation is slightly lower (−0.006); blur does not hurt (−0.002). The gap does not look photometric; M2 intervals are too wide to settle it.
- **P2, where to adapt:** LoRA adds +0.005 to +0.007 over the probe on M1 in every zone, and the zones do not differ (late − early: −0.000, [−0.006, +0.015]). Adaptation adds little wherever it goes.
- **P3, errors:** mIoU is 0.862 where three experts agree and 0.546 where one labelled (+0.315, [+0.180, +0.413]). Calibration is good overall (ECE 0.029), but on pixels predicted big rock confidence exceeds accuracy by 0.674 [+0.435, +0.762], and 45 % of big-rock pixels are missed.

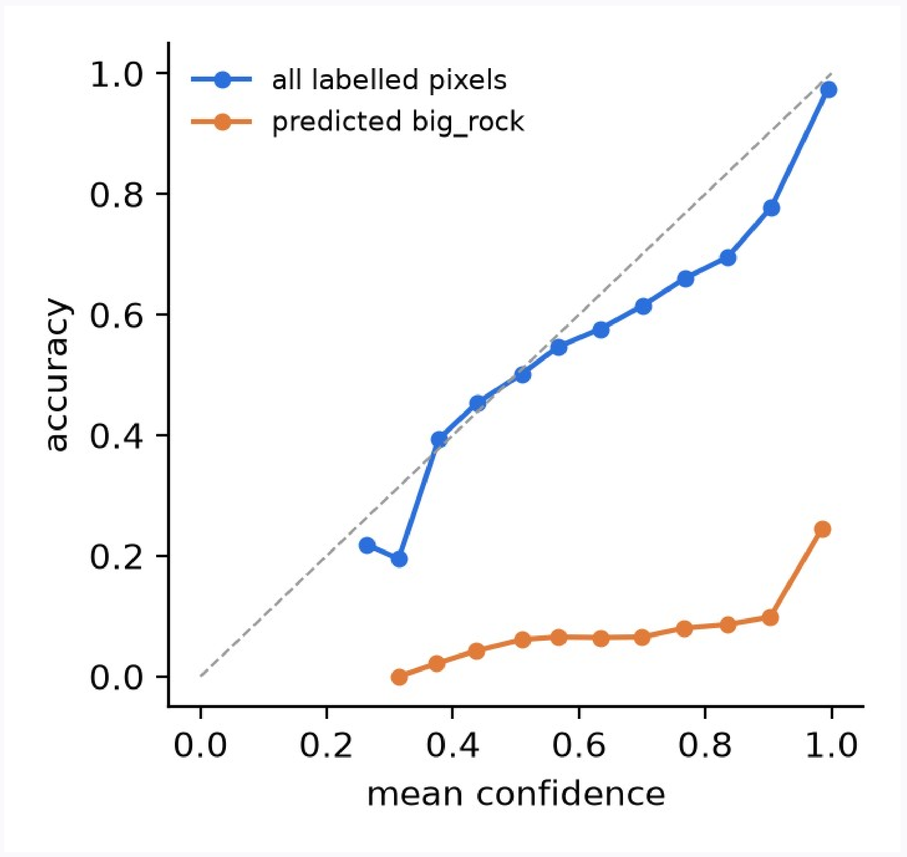


Fig. 2. P3: reliability of the probe on all labelled pixels and on pixels predicted as big rock.

Test mIoU (M1 / M2), trainable parameters:
Probe, frozen DINOv3 (3 seeds) — 4.6 k — 0.674 / 0.723
+ LoRA early (1–2) — 221 k — 0.681 / 0.730
+ LoRA middle (6–7) — 221 k — 0.679 / 0.728
+ LoRA late (11–12) — 221 k — 0.681 / 0.731
DeepLabV3-R50, ImageNet — 39.6 M — 0.667 / 0.713

CONTINUE THE CONVERSATION

Code, notebook and results: https://github.com/gonzaquis29/DinoV3-AI4Mars

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Data and model: AI4Mars (NASA/JPL), DINOv3 (Meta AI). Compute: free-tier Kaggle GPU.

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WAIMLAp 2026 • November 18–20, 2026 • Lima, Peru • waimlap.org